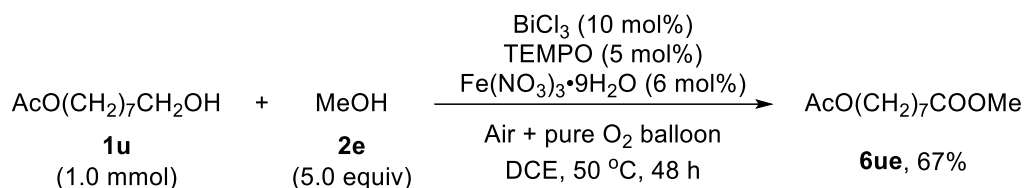
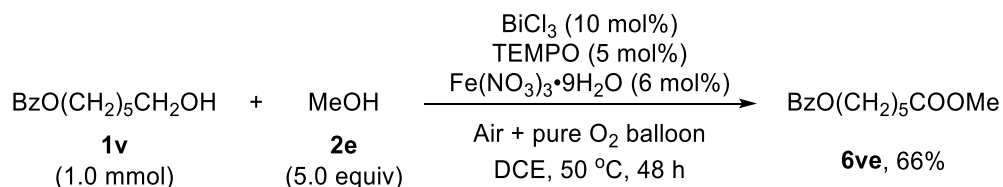


(21) Preparation of methyl 8-acetyloxyoctanoate (**6ue**, yyb-7-076)



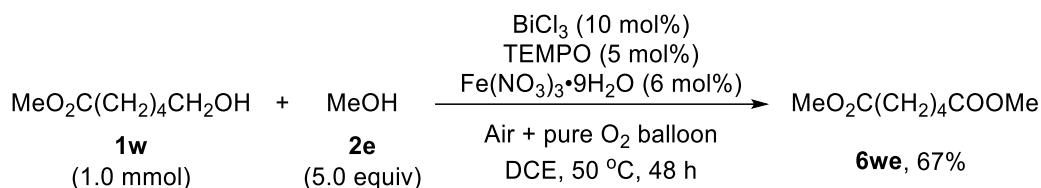
Following Typical Procedure II, the reaction of **1u** (189.1 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.3 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.4 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ue** (144.7 mg, 67%) [eluent: petroleum ether/ethyl acetate = 15/1 (~480 mL) to 10/1 (~110 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 4.05 (t, *J* = 6.8 Hz, 2 H, OCH₂), 3.67 (s, 3 H, OCH₃), 2.31 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.04 (s, 3 H, CH₃), 1.70-1.56 (m, 4 H, 2 x CH₂), 1.43-1.28 (m, 6 H, 3 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.0, 171.0, 64.3, 51.3, 33.8, 28.8, 28.7, 28.4, 25.6, 24.7, 20.8; IR (neat, cm⁻¹): ν = 2937, 2858, 1735, 1437, 1365, 1234, 1170, 1034; MS (ESI) *m/z*: 239 (M+Na)⁺, 217 (M+H)⁺; HRMS (ESI) calcd *m/z* for C₁₁H₂₁O₄ [M+H]⁺: 217.1434, found 217.1429.

(22) Preparation of 6-methoxy-6-oxohexyl benzoate (**6ve**, yyb-7-077)



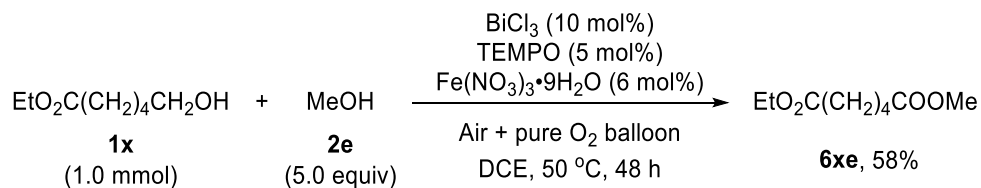
Following Typical Procedure II, the reaction of **1v** (220.1 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.5 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl₃ (31.0 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ve** (164.3 mg, 66%) [eluent: petroleum ether/ethyl acetate = 20/1 (~520 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 8.04 (d, *J* = 7.2 Hz, 2 H, Ar-H), 7.54 (t, *J* = 7.4 Hz, 1 H, Ar-H), 7.43 (t, *J* = 7.6 Hz, 2 H, Ar-H), 4.32 (t, *J* = 6.6 Hz, 2 H, OCH₂), 3.66 (s, 3 H, OCH₃), 2.34 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.79 (quint, *J* = 7.1 Hz, 2 H, CH₂), 1.71 (quint, *J* = 7.7 Hz, 2 H, CH₂), 1.54-1.42 (m, 2 H, CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.7, 166.4, 132.7, 130.2, 129.3, 128.2, 64.5, 51.3, 33.7, 28.2, 25.4, 24.4; IR (neat, cm⁻¹): ν = 2951, 1735, 1715, 1451, 1436, 1271, 1171, 1114; MS (70 eV, EI) *m/z* (%): 250 (M⁺, 1.09), 105 (100); HRMS (EI) calcd *m/z* for C₁₄H₁₈O₄ [M]⁺: 250.1200, Found: 250.1204.

(23) Preparation of dimethyl adipate (**6we**, wj-3-163)



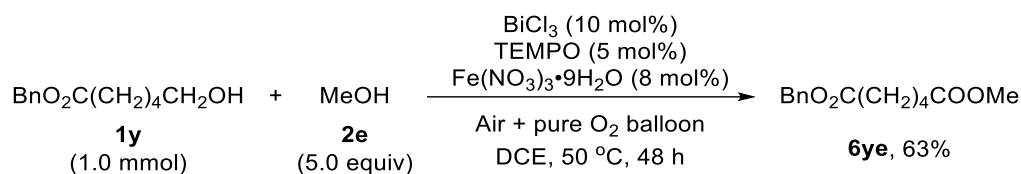
Following Typical Procedure I, the reaction of **1w** (146.3 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.4 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.4 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6we**¹¹ (116.3 mg, 67%) [eluent: petroleum ether/DCM = 2/1 (~300 mL) to 1/1 (~400 mL), then DCM (400 mL)] as oil. ¹H NMR (400 MHz, CDCl_3): δ = 3.56 (s, 6 H, 2 x OCH_3), 2.27-2.13 (m, 4 H, 2 x CH_2), 1.60-1.47 (m, 4 H, 2 x CH_2); ¹³C NMR (100 MHz, CDCl_3) δ = 173.7, 51.5, 33.6, 24.3; IR (neat, cm^{-1}) ν = 2955, 1734, 1437, 1367, 1248, 1197, 1171, 1083; MS (70 eV, EI) m/z (%): 143 ($(\text{M}-\text{OMe})^+$, 73.02), 114 (100).

(24) Preparation of ethyl methyl adipate (**6xe**, yyb-7-039)



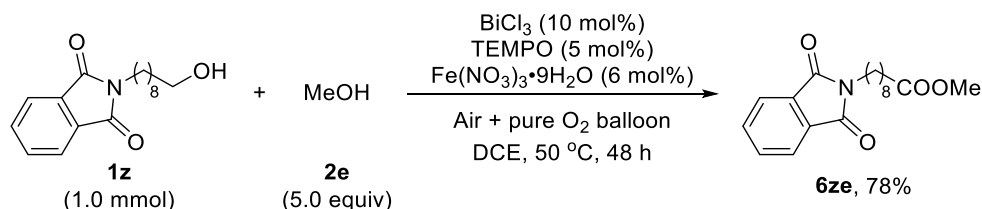
Following Typical Procedure II, the reaction of **1x** (160.7 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.8 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl_3 (32.0 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6xe**¹² (110.3 mg, 58%) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL) to 10/1 (~330 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl_3): δ = 4.13 (q, J = 7.1 Hz, 2 H, OCH_2), 3.67 (s, 3 H, OCH_3), 2.40-2.26 (m, 4 H, 2 x CH_2), 1.73-1.58 (m, 4 H, 2 x CH_2), 1.26 (t, J = 7.2 Hz, 3 H, CH_3); ¹³C NMR (100 MHz, CDCl_3) δ = 173.7, 173.2, 60.2, 51.4, 33.8, 33.5, 24.3, 24.2, 14.1; IR (neat, cm^{-1}) ν = 2981, 2945, 2873, 1731, 1438, 1372, 1242, 1172, 1143; MS (ESI) m/z : 211 ($(\text{M}+\text{Na})^+$, 189 ($(\text{M}+\text{H})^+$.

(25) Preparation of benzyl methyl adipate (**6ye**, yyb-7-036)



Following Typical Procedure II, the reaction of **1y** (223.8 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (32.3 mg, 0.08 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.6 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ye**¹³ (157.9 mg, 63%) [eluent: petroleum ether/ethyl acetate = 25/1 (~260 mL) to 20/1 (~630 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.44-7.25 (m, 5 H, Ar-H), 5.11 (s, 2 H, OCH₂), 3.65 (s, 3 H, OCH₃), 2.37 (t, *J* = 7.0 Hz, 2 H, CH₂), 2.32 (t, *J* = 7.0 Hz, 2 H, CH₂), 1.76-1.58 (m, 4 H, 2 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.6, 173.0, 135.9, 128.4, 128.1, 66.0, 51.4, 33.7, 33.5, 24.2; IR (neat, cm⁻¹): ν = 2951, 1731, 1455, 1438, 1381, 1358, 1164, 1140; MS (ESI) *m/z*: 273 (M+Na)⁺, 251 (M+H)⁺.

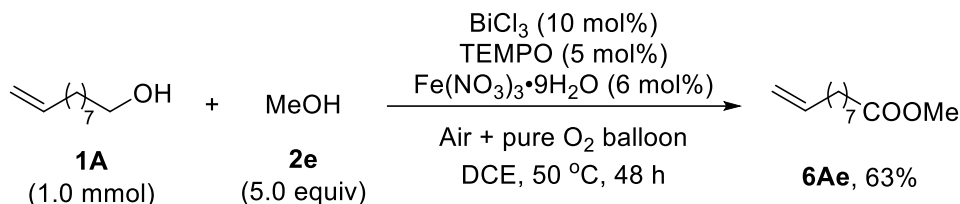
(26) Preparation of methyl 9-(1,3-dioxoisindolin-2-yl)nonanoate (**6ze**, yyb-6-197)



Following Typical Procedure I, the reaction of **1z** (290.2 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.8 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ze** (248.2 mg, 78%) [eluent: petroleum ether/DCM = 3/1 (~200 mL) to 1/1 (~400 mL), then 1/2 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.91-7.78 (m, 2 H, Ar-H), 7.77-7.65 (m, 2 H, Ar-H), 3.76-3.55 (m, 5 H, OCH₃ and NCH₂), 2.29 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.78-1.55 (m, 4 H, 2 x CH₂), 1.40-1.19 (m, 8 H, 4 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.9, 168.1, 133.6, 131.9, 122.9, 51.1, 37.7, 33.7, 28.8, 28.74, 28.69, 28.3, 26.5, 24.6; IR (neat, cm⁻¹): ν = 2928, 2855, 1736, 1707, 1466, 1436, 1395, 1170, 1060; MS (70 eV, EI) *m/z* (%): 317 (M⁺, 9.61), 160 (100); HRMS (EI) calcd *m/z* for

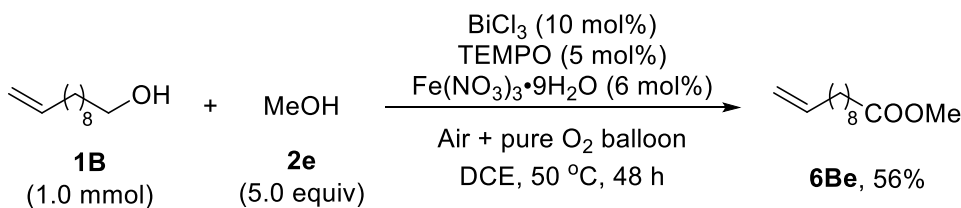
C₁₈H₂₃NO₄ [M]⁺: 317.1622, Found: 317.1623.

(27) Preparation of methyl dec-9-enoate (6Ae, yyb-7-075)



Following Typical Procedure II, the reaction of **1A** (165.1 mg, 95% purity, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6Ae**¹⁴ (118.6 mg, 63%, 98% purity) [eluent: petroleum ether/ethyl acetate = 60/1 (~300 mL) to 50/1 (~250 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 5.87-5.73 (m, 1 H, =CH), 5.03-4.95 (m, 1 H, one proton of =CH₂), 4.95-4.88 (m, 1 H, one proton of =CH₂), 3.66 (s, 3 H, OCH₃), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.04 (q, *J* = 6.9 Hz, 2 H, CH₂), 1.67-1.57 (m, 2 H, CH₂), 1.42-1.25 (m, 8 H, 4 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.2, 139.0, 114.1, 51.3, 34.0, 33.7, 29.0, 28.83, 28.76, 24.9; IR (neat, cm⁻¹): ν = 3080, 2927, 2855, 1740, 1463, 1436, 1361, 1198, 1169; MS (ESI) *m/z*: 185 (M+H)⁺.

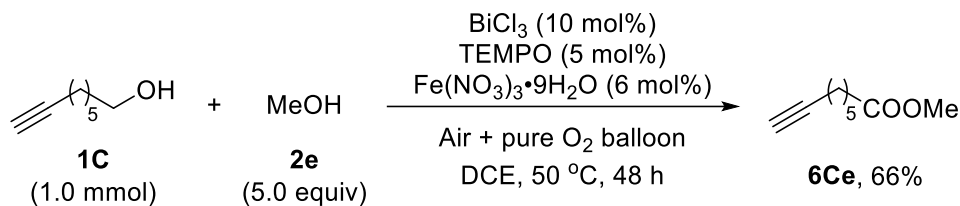
(28) Preparation of methyl undec-10-enoate (6Be, yyb-6-147)



Following Typical Procedure II, the reaction of **1B** (171.3 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.4 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.3 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6Be**¹⁴ (115.9 mg, 56%, 96% purity) [eluent: petroleum ether/ethyl acetate = 50/1 (~250 mL)] as yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 5.87-5.73 (m, 1 H, =CH), 5.04-4.95 (m, 1 H, one proton of =CH₂), 4.95-4.88 (m, 1 H, one proton of =CH₂), 3.66 (s, 3 H, OCH₃), 2.30 (t,

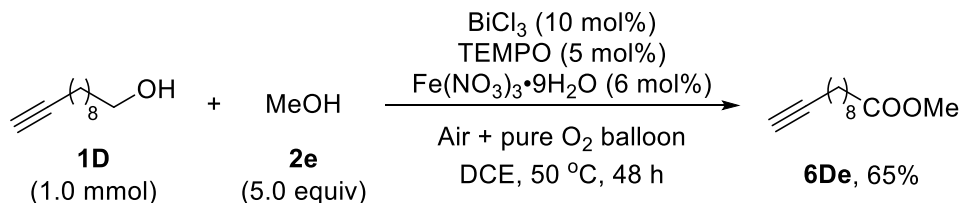
$J = 7.4$ Hz, 2 H, CH₂), 2.10-1.97 (m, 2 H, CH₂), 1.67-1.57 (m, 2 H, CH₂), 1.42-1.20 (m, 10 H, 5 x CH₂); ¹³C NMR (100 MHz, CDCl₃): $\delta = 174.2, 139.1, 114.1, 51.3, 34.0, 33.7, 29.2, 29.1, 29.05, 28.97, 28.8, 24.9$; IR (neat, cm⁻¹): $\nu = 3077, 2926, 2855, 1740, 1463, 1436, 1361, 1197, 1170$; MS (ESI) m/z : 199 (M+H)⁺.

(29) Preparation of methyl hept-6-ynoate (**6Ce**, yyb-7-074)



Following Typical Procedure II, the reaction of **1C** (126.9 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.5 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.8 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Ce**¹⁵ (103.9 mg, 66%, 98% purity) [eluent: petroleum ether/ethyl acetate = 30/1 (~460 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): $\delta = 3.67$ (s, 3 H, OCH₃), 2.33 (t, $J = 7.6$ Hz, 2 H, CH₂), 2.20 (td, $J_1 = 7.0$ Hz, $J_2 = 2.7$ Hz, 2 H, CH₂), 1.95 (t, $J = 2.6$ Hz, 1 H, CH), 1.65 (quint, $J = 7.5$ Hz, 2 H, CH₂), 1.60-1.50 (m, 2 H, CH₂), 1.48-1.38 (m, 2 H, CH₂); ¹³C NMR (100 MHz, CDCl₃): $\delta = 174.0, 84.2, 68.3, 51.4, 33.8, 28.1, 28.0, 24.3, 18.1$; IR (neat, cm⁻¹): $\nu = 3295, 2940, 2863, 2116, 1735, 1460, 1436, 1364, 1202, 1172$; MS (ESI) m/z : 155 (M+H)⁺.

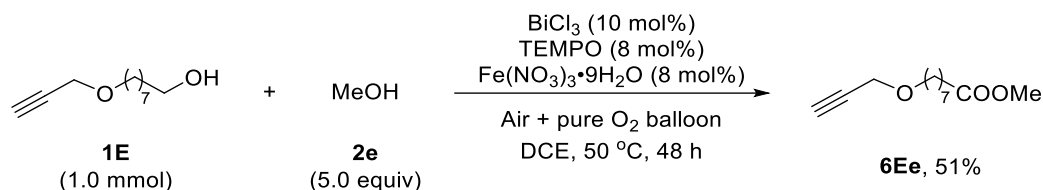
(30) Preparation of methyl undec-10-ynoate (**6De**, yyb-6-006)



Following Typical Procedure I, the reaction of **1D** (167.8 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (32.1 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6De**¹⁶ (127.6 mg, 65%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1

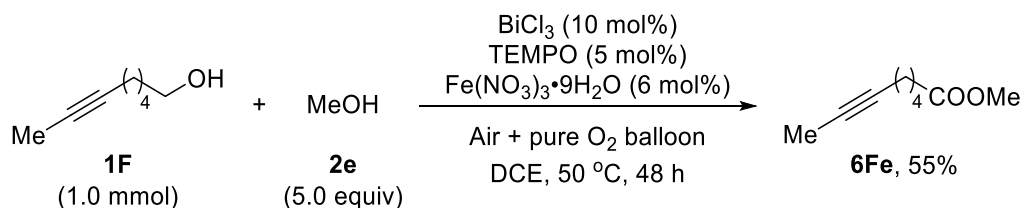
(~300 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.66 (s, 3 H, OCH₃), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.17 (t, *J* = 7.0 Hz, 2 H, CH₂), 1.94 (s, 1 H, CH), 1.68-1.57 (m, 2 H, CH₂), 1.52 (quint, *J* = 7.2 Hz, 2 H, CH₂), 1.45-1.24 (m, 8 H, 4 x CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 174.1, 84.5, 68.0, 51.3, 33.9, 29.0, 28.9, 28.8, 28.5, 28.3, 24.8, 18.2; **IR** (neat, cm⁻¹): ν = 3304, 2933, 2857, 2116, 1737, 1459, 1438, 1361, 1197, 1170; **MS** (ESI) *m/z*: 219 (M+Na)⁺, 197 (M+H)⁺.

(31) Preparation of methyl 8-(prop-2-ynoxy)octanoate (**6Ee**, yyb-7-081)



Following Typical Procedure II, the reaction of **1E** (182.7 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (31.9 mg, 0.08 mmol), TEMPO (12.7 mg, 0.08 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6Ee** (108.3 mg, 51%) [eluent: petroleum ether/ ethyl acetate = 15/1 (~320 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 4.13 (d, *J* = 2.4 Hz, 2 H, OCH₂), 3.66 (s, 3 H, OCH₃), 3.50 (t, *J* = 6.4 Hz, 2 H, OCH₂), 2.43 (t, *J* = 2.4 Hz, 1 H, CH), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.70-1.52 (m, 4 H, 2 x CH₂), 1.42-1.26 (m, 6 H, 3 x CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 174.1, 79.9, 74.0, 70.0, 57.9, 51.3, 33.9, 29.3, 28.92, 28.90, 25.8, 24.7; **IR** (neat, cm⁻¹): ν = 3274, 2932, 2857, 2115, 1735, 1437, 1357, 1249, 1171, 1098; **MS** (ESI) *m/z*: 235 (M+Na)⁺, 213 (M+H)⁺; **HRMS** (ESI) calcd *m/z* for C₁₂H₂₁O₃ [M+H]⁺: 213.1485, found 213.1484.

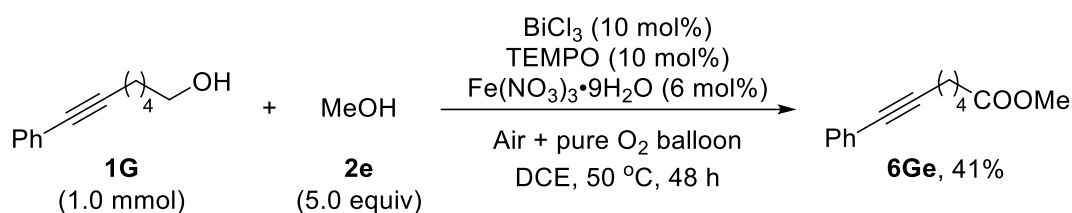
(32) Preparation of methyl 7-methylhept-6-ynoate (**6Fe**, zhangq-3-158)



Following Typical Procedure I, the reaction of **1F** (126.0 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.9 mg, 0.06 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃ (31.7 mg,

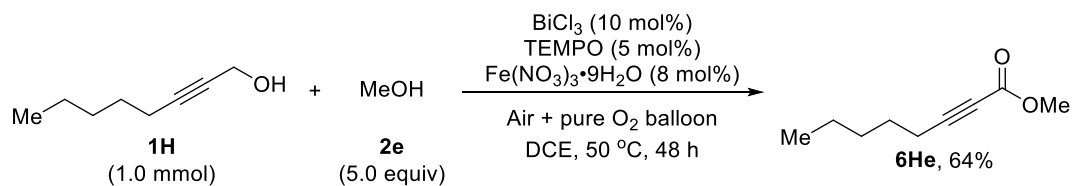
0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Fe**¹⁷ (85.3 mg, 55%) [eluent: petroleum ether/DCM = 10/1 (~440 mL) to 4/1 (~240 mL), then 2/1 (~150 mL)] as oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 2.33 (t, J = 7.6 Hz, 2 H, CH₂), 2.20-2.06 (m, 2 H, CH₂), 1.77 (s, 3 H, CH₃), 1.75-1.65 (m, 2 H, CH₂), 1.55-1.45 (m, 2 H, CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 173.9, 78.5, 75.7, 51.3, 33.5, 28.3, 24.0, 18.3, 3.3; **IR** (neat, cm⁻¹): ν = 2946, 2924, 2220, 1736, 1437, 1358, 1201, 1172, 1152; **MS** (ESI) m/z : 177 (M+Na)⁺, 155 (M+H)⁺.

(33) Preparation of methyl 7-phenylhept-6-ynoate (**6Ge**, yyb-7-071)



Following Typical Procedure II, the reaction of **1G** (189.1 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.8 mg, 0.06 mmol), TEMPO (16.2 mg, 0.1 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Ge**¹⁸ (90.7 mg, 41%, 98% purity) [eluent: petroleum ether/ethyl acetate = 50/1 (~510 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 7.45-7.35 (m, 2 H, Ar-H), 7.31-7.22 (m, 3 H, Ar-H), 3.67 (s, 3 H, OCH₃), 2.43 (t, J = 7.0 Hz, 2 H, CH₂), 2.37 (t, J = 7.4 Hz, 2 H, CH₂), 1.87-1.75 (m, 2 H, CH₂), 1.69-1.59 (m, 2 H, CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 173.9, 131.5, 128.1, 127.5, 123.8, 89.5, 80.9, 51.5, 33.5, 28.1, 24.1, 19.1; **IR** (neat, cm⁻¹): ν = 2948, 2232, 1735, 1490, 1437, 1359, 1199, 1172, 1146; **MS** (70 eV, EI) m/z (%): 216 (M⁺, 28.56), 115 (100).

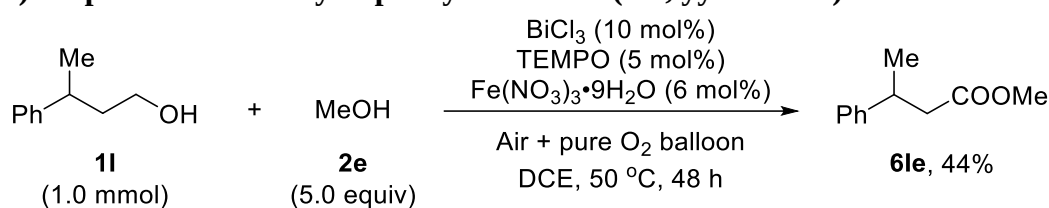
(34) Preparation of methyl oct-2-ynoate (**6He**, yyb-11-173A)



Following Typical Procedure II, the reaction of **1H** (127.7 mg, 99% purity, 1.0 mmol), Fe(NO₃)₃·9H₂O (32.6 mg, 0.08 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃

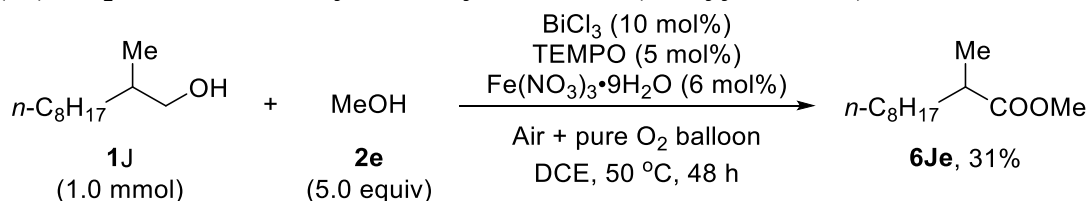
(31.3 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6He**¹⁹ (99.3 mg, 64%) [eluent: petroleum ether/DCM = 10/1 (~270 mL) to 5/1 (~240 mL), then 3/1 (~200 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.76 (s, 3 H, OCH₃), 2.33 (t, J = 7.0 Hz, 2 H, CH₂), 1.59 (quint, J = 7.2 Hz, 2 H, CH₂), 1.44-1.24 (m, 4 H, 2 x CH₂), 0.91 (t, J = 7.0 Hz, 3 H, CH₃); **¹³C NMR** (100 MHz, CDCl₃): δ = 154.2, 89.8, 72.7, 52.4, 30.8, 27.1, 22.0, 18.5, 13.7; **IR** (neat, cm⁻¹): ν = 2956, 2863, 2236, 1713, 1458, 1434, 1247, 1075; **MS** (ESI) m/z : 155 (M+H)⁺.

(35) Preparation of methyl 3-phenylbutanoate (**6Ie**, yyb-12-158)



Following Typical Procedure II, the reaction of **1I** (150.0 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.3 mg, 0.06 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃ (31.3 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Ie**²⁰ (77.6 mg, 44%) [eluent: petroleum ether/ethyl acetate = 60/1 (~300 mL) to 40/1 (~200 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 7.34-7.26 (m, 2 H, Ar-H), 7.25-7.15 (m, 3 H, Ar-H), 3.62 (s, 3 H, OCH₃), 3.36-3.20 (m, 1 H, CH), 2.63 (dd, J_1 = 15.2 Hz, J_2 = 6.8 Hz, 1 H, one proton of CH₂), 2.54 (dd, J_1 = 15.0 Hz, J_2 = 8.2 Hz, 1 H, one proton of CH₂), 1.30 (d, J = 6.8 Hz, 3 H, CH₃); **¹³C NMR** (100 MHz, CDCl₃): δ = 172.8, 145.7, 128.5, 126.7, 126.4, 51.4, 42.7, 36.4, 21.7; **IR** (neat, cm⁻¹): ν = 3029, 2964, 2930, 1735, 1495, 1436, 1378, 1290, 1164, 1085, 1022; **MS** (ESI) m/z : 179 (M+H)⁺.

(36) Preparation of methyl 2-methyldecanoate (**6Je**, yyb-12-157)



Following Typical Procedure II, the reaction of **1J** (172.8 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.7 mg, 0.06 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Je**²¹ (62.3 mg, 31%)

[eluent: petroleum ether/ethyl acetate = 50/1 (~410 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 2.50-2.37 (m, 1 H, CH), 1.71-1.57 (m, 1 H, one proton of CH₂), 1.46-1.35 (m, 1 H, one proton of CH₂), 1.35-1.17 (m, 12 H, 6 x CH₂), 1.14 (d, *J* = 6.8 Hz, 3 H, CH₃), 0.88 (t, *J* = 6.6 Hz, 3 H, CH₃); **¹³C NMR** (100 MHz, CDCl₃): δ = 177.4, 51.4, 39.4, 33.8, 31.8, 29.5, 29.4, 29.2, 27.2, 22.6, 17.0, 14.0; **IR** (neat, cm⁻¹): ν = 2953, 2925, 2855, 1738, 1463, 1434, 1378, 1195, 1164; **MS** (ESI) *m/z*: 201 (M+H)⁺.

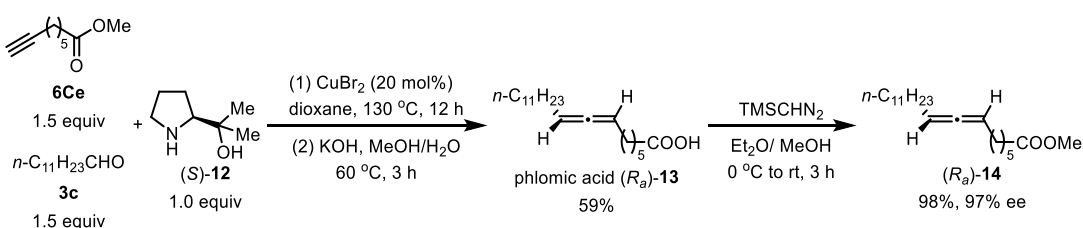
Total synthesis of phlomic acid:

(1) Preparation of methyl hept-6-ynoate (**6Ce**, yyb-7-156A)



Following Typical Procedure II, the reaction of **1C** (631.8 mg, 5.0 mmol), Fe(NO₃)₃·9H₂O (121.6 mg, 0.3 mmol), TEMPO (40.0 mg, 0.25 mmol), BiCl₃ (157.1 mg, 0.5 mmol), MgSO₄ (249.6 mg, 2.1 mmol), and MeOH (0.6 mL, 15 mmol) in DCE (15 mL) afforded **6Ce**¹⁵ (516.2 mg, 65%, 97% purity) [eluent: petroleum ether/ethyl acetate = 49/1 (~600 mL) to 24/1 (~300 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 2.33 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.25-2.13 (m, 2 H, CH₂), 1.99-1.91 (m, 1 H, CH), 1.65 (quint, *J* = 7.3 Hz, 2 H, CH₂), 1.55 (quint, *J* = 7.0 Hz, 2 H, CH₂), 1.50-1.38 (m, 2 H, CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 174.0, 84.2, 68.3, 51.4, 33.8, 28.1, 28.0, 24.3, 18.1.

(2) Preparation of (*R*)-7,8-Eicosadienoic acid (phlomic acid) ((*R_a*)-**13**, yyb-7-158, yyb-7-160, yyb-7-163)



To an oven dried screw cap reaction tube were added CuBr₂ (90.6 mg, 0.4 mmol),

(*S*)-**12** (271.6 mg, 95% purity, 2.0 mmol), **6Ce** (464.3 mg, 3.0 mmol)/dioxane (3 mL), and **3c** (573.4 mg, 97% purity, 3.0 mmol)/dioxane (3 mL) sequentially. The resulting mixture was stirred at 130 °C for 12 h as monitored by TLC, diluted with 60 mL of ethyl ether, washed with an aqueous solution of HCl (3 M, 40 mL), extracted with ethyl ether (20 mL x 3), washed with brine, and dried over anhydrous Na₂SO₄. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford (*R_a*)-**14** (397.4 mg) [eluent: petroleum ether/ethyl acetate = 99/1 (~1200 mL)] as light yellow oil.

To a round bottom flask were added KOH (198.3 mg, 3.0 mmol, 85% purity), (*R_a*)-**14** (397.4 mg), and MeOH/H₂O (12 mL, 4:1/v:v) sequentially. The resulting mixture was stirred at 60 °C for 3 h as monitored by TLC. HCl (3 M, 1.2 mL) was added dropwise at 0 °C with an ice-water bath. The resulting mixture was evaporated to remove MeOH, treated with DCM (30 mL) and H₂O (25 mL), extracted with DCM (20 mL x 3), washed with brine, and dried over anhydrous Na₂SO₄. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford (*R_a*)-**13**¹⁵ (361.8 mg, 59%) [eluent: petroleum ether/ethyl acetate = 10/1 (~440 mL) to 4/1 (~370 mL)] as light yellow solid: m.p. 30.4-31.5 °C (We were not able to recrystallize from the solvents tested, the m.p. of (*R_a*)-**13** was determined by using the solid right after evaporation of the eluent). $[\alpha]_{\text{D}}^{27} = -41.3$ (c = 0.94, CHCl₃) (reported:¹⁵ $[\alpha]_{\text{D}}^{30.5} = -40.7$ (c = 1.03, CHCl₃)); ¹H NMR (400 MHz, CDCl₃) δ = 5.12-5.01 (m, 2 H, CH=C=CH), 2.35 (t, *J* = 7.6 Hz, 2 H, CH₂), 2.04-1.92 (m, 4 H, 2 x CH₂), 1.65 (quint, *J* = 7.3 Hz, 2 H, CH₂), 1.48-1.18 (m, 22 H, 11 x CH₂), 0.88 (t, *J* = 6.6 Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃): δ = 203.9, 180.5, 91.1, 90.5, 34.1, 31.9, 29.6, 29.5, 29.4, 29.2, 29.1, 29.0, 28.71, 28.68, 28.5, 24.5, 22.7, 14.1.

The ee of (*R_a*)-**13** was determined by HPLC analysis after being converted to (*R_a*)-**14**: To the flask containing (*R_a*)-**13** (62.0 mg, 0.2 mmol) was added ethyl ether/MeOH (5.0 mL, 4:1/v:v). TMSCHN₂ (0.15 mL, 2.0 M in hexane, 0.3 mmol) was added dropwise at 0 °C. The resulting mixture was warmed up to room temperature and then stirred for 3 h as monitored by TLC. The solvent was removed under reduced pressure

and the residue was purified by chromatography on silica gel to afford (*R_a*)-**14**¹⁵ (63.5 mg, 98%, 97% ee) [eluent: petroleum ether/ethyl acetate = 100/1 (~200 mL) to 20/1 (~210 mL)] as colorless oil; HPLC conditions: LC column, hexane, 1.5 mL/min, λ = 214 nm, t_R (major) = 8.1 min, t_R (minor) = 9.8 min; $[\alpha]_D^{27}$ = -39.3 (c = 1.075, CHCl₃) (reported:¹⁵ $[\alpha]_D^{30.5}$ = -39.9 (c = 0.99, CHCl₃)); **¹H NMR** (400 MHz, CDCl₃) δ = 5.16-4.98 (m, 2 H, CH=C=CH), 3.66 (s, 3 H, OCH₃), 2.31 (t, J = 7.4 Hz, 2 H, CH₂), 2.05-1.90 (m, 4 H, 2 x CH₂), 1.63 (quint, J = 7.5 Hz, 2 H, CH₂), 1.48-1.16 (m, 22 H, 11 x CH₂), 0.88 (t, J = 6.6 Hz, 3 H, CH₃); **¹³C NMR** (100 MHz, CDCl₃): δ = 203.8, 174.2, 91.1, 90.5, 51.4, 34.0, 31.9, 29.65, 29.63, 29.62, 29.5, 29.3, 29.2, 29.1, 29.0, 28.73, 28.72, 28.6, 24.8, 22.7, 14.1.

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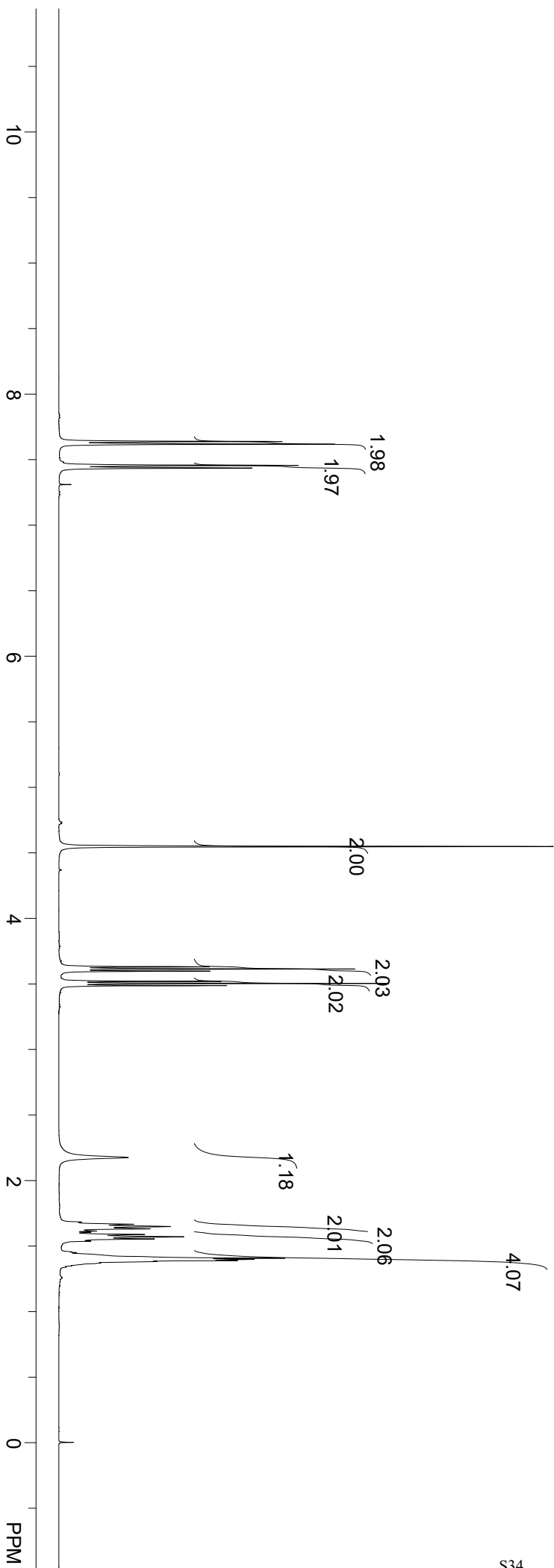
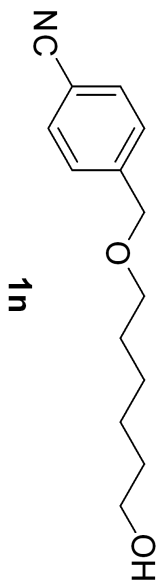
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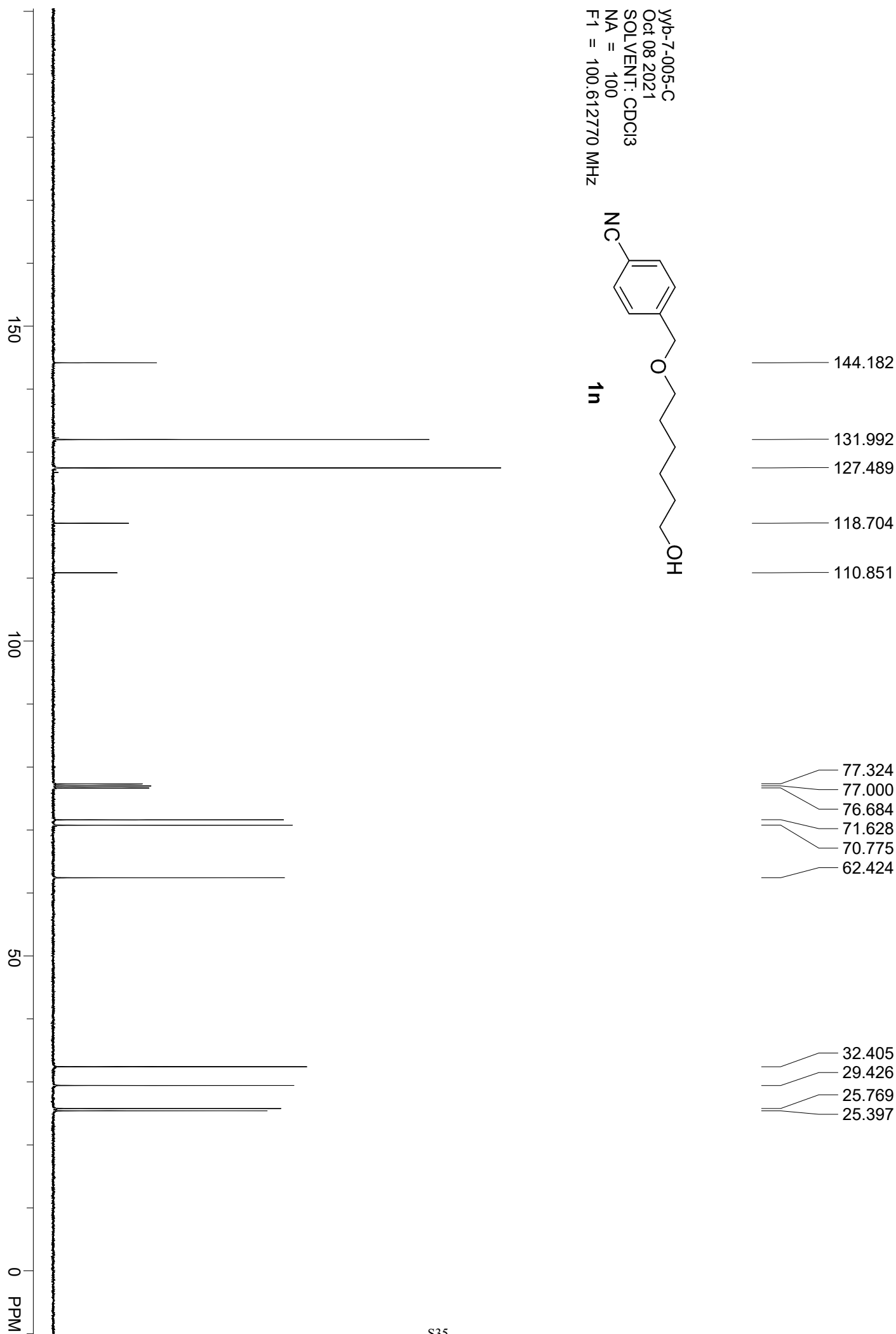
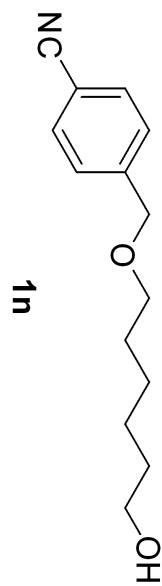
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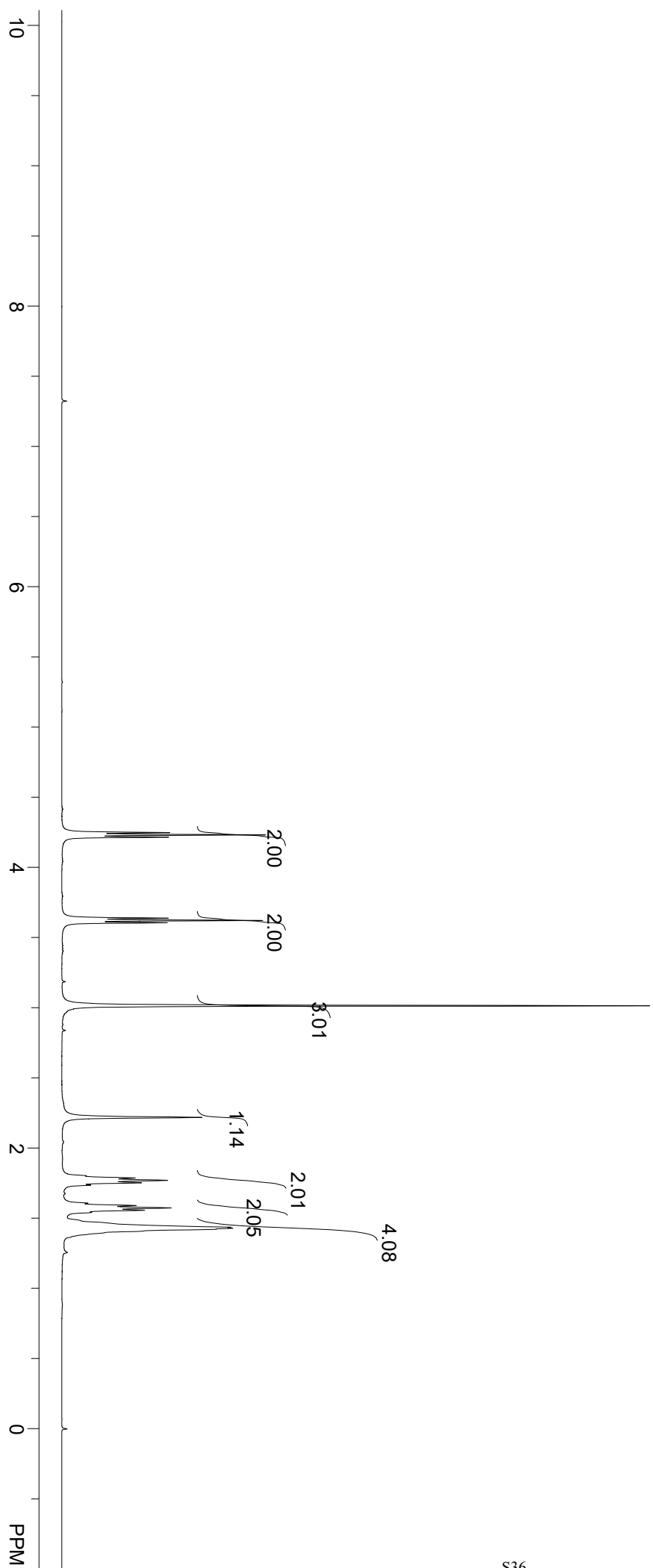
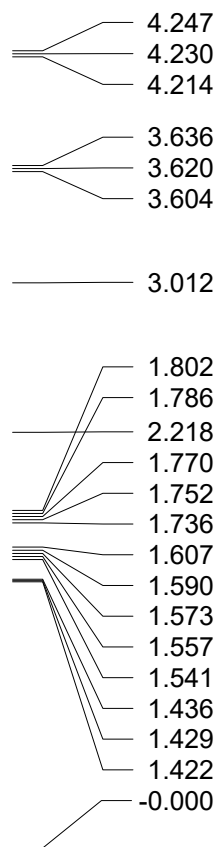
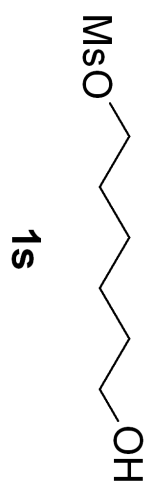
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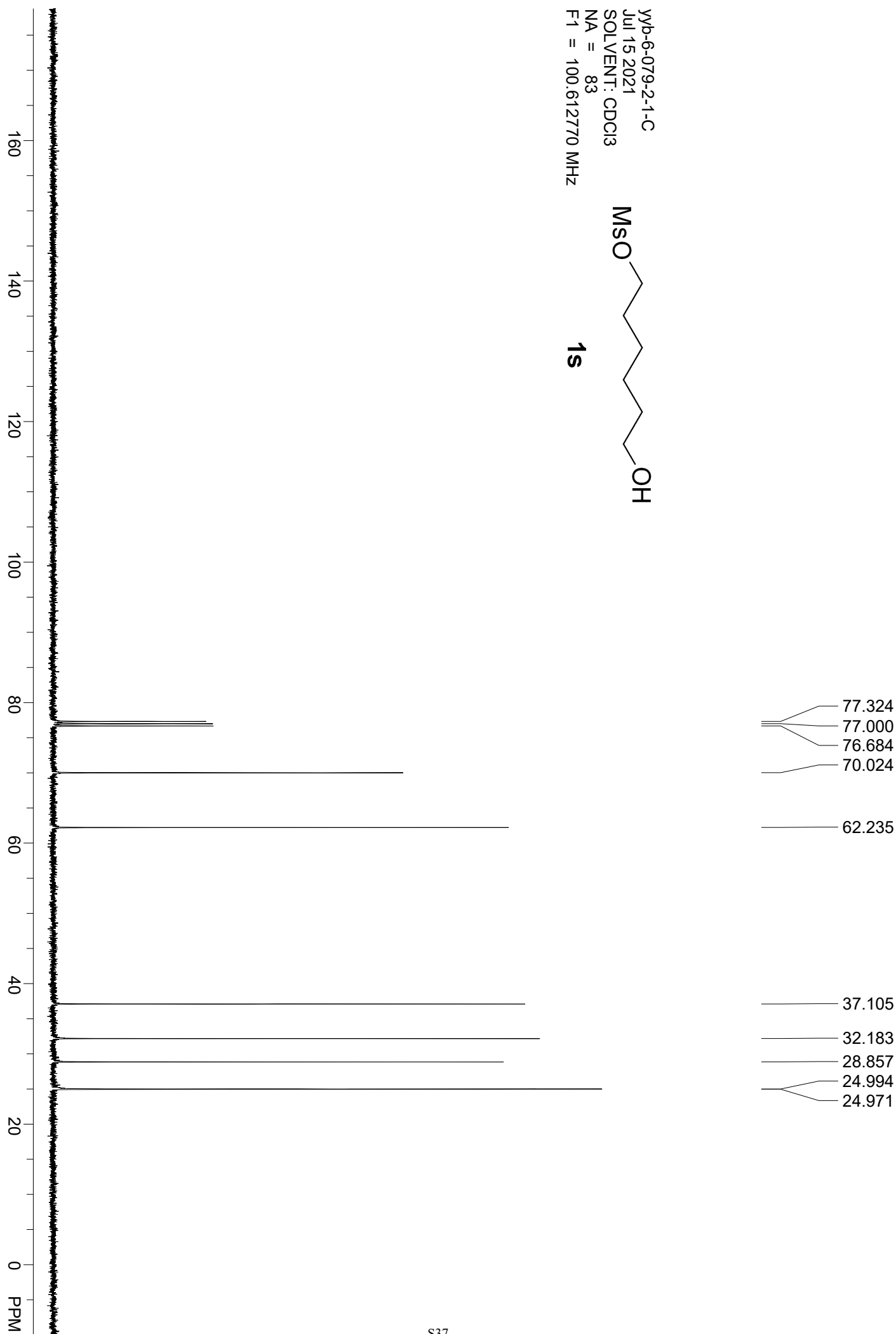
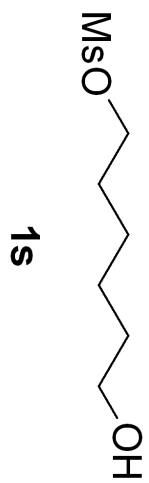
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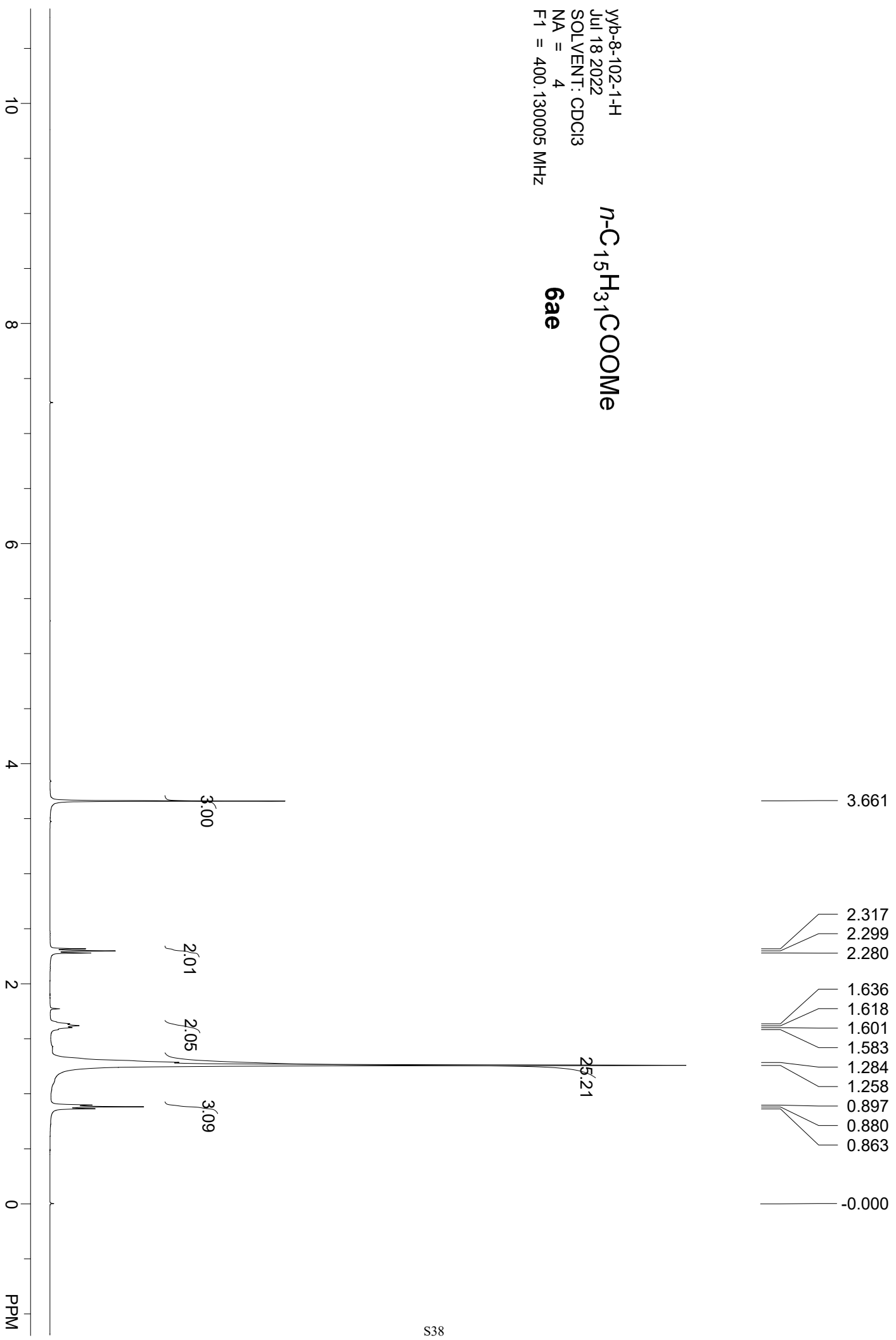
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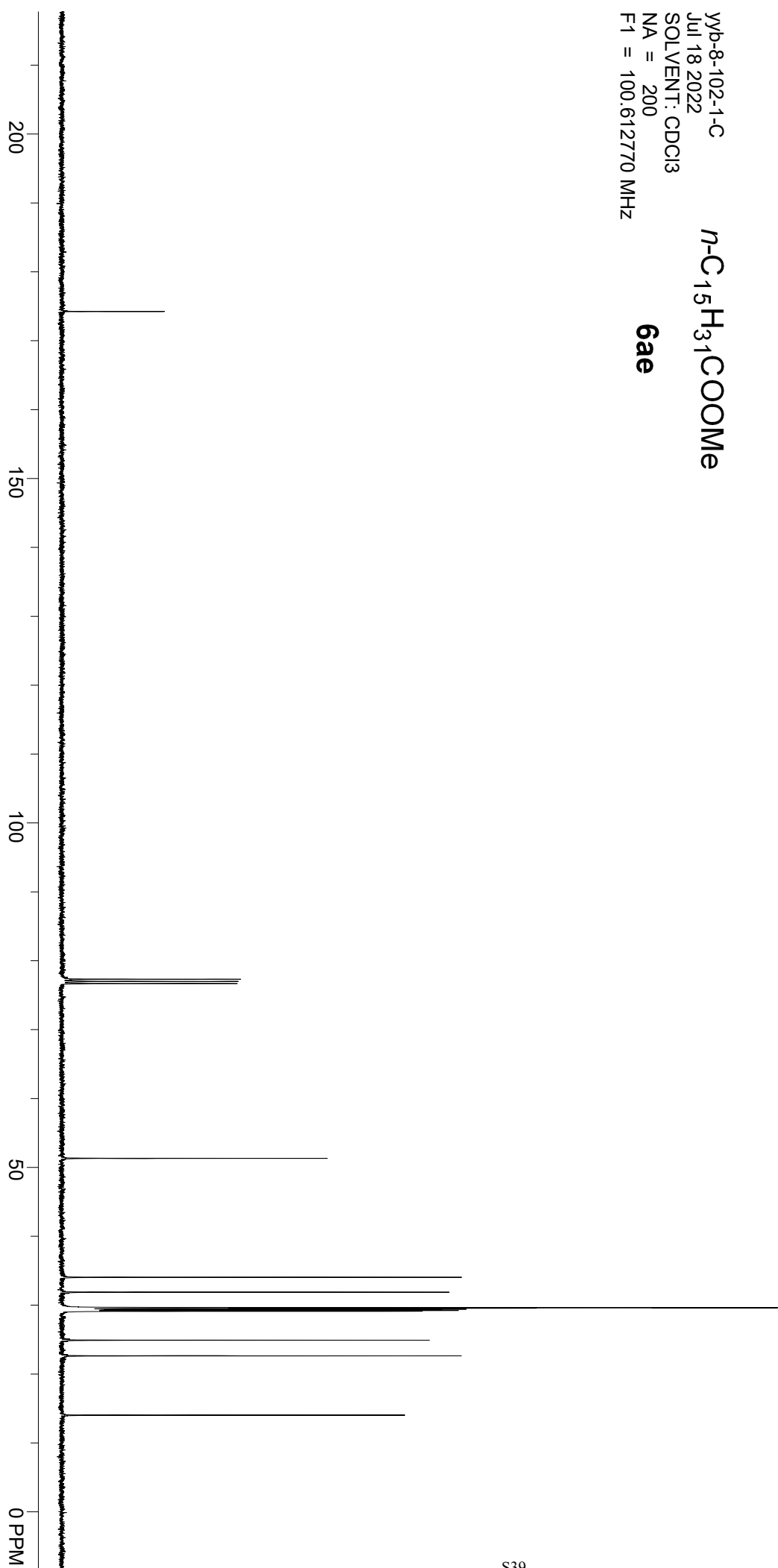


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